2 Lines in the plane

Sheet 5 Mon 2 Nov. 2026 *Direction vector of a line*

1. Warm-up · Quick [GEO] Give a vector collinear to $\vec{u}\begin{pmatrix}2\\-4\end{pmatrix}$ with simpler integer coordinates.

   Ans. $\begin{pmatrix}1\\-2\end{pmatrix}$ (divide by 2).

2. Core · Method [GEO] Give a direction vector of (AB) for $A(1,2)$ and $B(4,8)$.

   Ans. $\overrightarrow{AB}\begin{pmatrix}3\\6\end{pmatrix}$, or $\begin{pmatrix}1\\2\end{pmatrix}$, which is collinear to it.

3. Core + · MCQ [GEO] A direction vector of the line through $A(0,1)$ and $B(2,-3)$ is:

   A. $\begin{pmatrix}1\\-2\end{pmatrix}$ **B.** $\begin{pmatrix}1\\-4\end{pmatrix}$ **C.** $\begin{pmatrix}-4\\-2\end{pmatrix}$ **D.** $\begin{pmatrix}2\\4\end{pmatrix}$

   Ans. A. $\begin{pmatrix}1\\-2\end{pmatrix}$

   Here $\overrightarrow{AB}\begin{pmatrix}2\\-4\end{pmatrix}$*; test collinearity with the determinant.*

4. Review · Quick [STA] Marks: 8, 12, 12, 15, 18. Calculate the mean, the median and the range.

   Ans. Mean 13; median 12; range 10.

5. Thread · Property [DEM] ABC is right-angled at A and I is the midpoint of $[BC]$. What can be said about IA, IB and IC? Why?

   Ans. $IA = IB = IC$: the midpoint of the hypotenuse is the centre of the circumcircle.

Sheet 6 Tue 3 Nov. 2026 *Cartesian equation of a line*

1. Warm-up · Quick [SUB] Calculate $3x - 2y + 1$ for $x = 1$ and $y = -2$.

   Ans. $3 + 4 + 1 = 8$.

2. Core · Method [GEO] The line d has equation $2x + 3y - 6 = 0$. Give a direction vector and a point of d.

   Ans. For $ax + by + c = 0$, $\begin{pmatrix}-b\\a\end{pmatrix}$ is a direction vector: $\begin{pmatrix}-3\\2\end{pmatrix}$. The point $(3,0)$ works: $6 + 0 - 6 = 0$.

3. Core + · True/false [GEO] True or false? Justify. The lines $2x + 3y - 6 = 0$ and $4x + 6y + 1 = 0$ are parallel.

   Ans. True. Direction vectors $\begin{pmatrix}-3\\2\end{pmatrix}$ and $\begin{pmatrix}-6\\4\end{pmatrix}$ are collinear; the lines are distinct: $(3,0)$ does not lie on the second line ($12 + 1 \neq 0$).

4. Review · Table [LEC] Mode of transport of 30 students.

Transport	Bus	Bike	Walking	Car
Frequency	12	5	8	5

   Give the proportion of students who come by bus and the angle of the "bike" sector in a pie chart.

   Ans. $\frac{12}{30} = 40\%$; $\frac{5}{30} \times 360° = 60°$.

5. Thread · Quick [REP] On a number line, A has coordinate -2 and B has coordinate 3. Give AB and the coordinate of the midpoint of $[AB]$.

   Ans. $AB = 3 - (-2) = 5$; midpoint: $\frac{-2+3}{2} = 0.5$.

Sheet 7 Thu 5 Nov. 2026 *Slope-intercept equation and gradient of a non-vertical line*

1. Warm-up · Quick [EQ] Make y the subject of $2x + y - 5 = 0$.

   Ans. $y = -2x + 5$.

2. Core · Method [GEO] Give the gradient and the y-intercept of the line $3x - 2y + 4 = 0$.

   Ans. $-2y = -3x - 4$, $y = \frac{3}{2}x + 2$: gradient $\frac{3}{2}$, y-intercept 2.
   Pitfall: divide every term by -2 (signs).

3. Core + · MCQ [GEO] The gradient of the line (AB) with $A(1,2)$ and $B(5,0)$ is:

   A. 2 **B.** $\frac{1}{2}$ **C.** -2 **D.** $-\frac{1}{2}$

   Ans. D. $-\frac{1}{2}$
   Gradient $= \frac{y_B - y_A}{x_B - x_A} = \frac{-2}{4}$.

4. Review · Quick [CAL] Calculate $\frac{3}{4} + \frac{5}{6}$.

   Ans. $\frac{9}{12} + \frac{10}{12} = \frac{19}{12}$.

5. Thread · Problem [DIV] Two gears have 12 and 18 teeth. After how many turns of each do they return together to their starting position?

   Ans. $\text{lcm}(12,18) = 36$ teeth: 3 turns of the first, 2 turns of the second.

Sheet 8 Mon 9 Nov. 2026 *Drawing a line given by an equation*

1. Warm-up · Quick [IMG] For $y = -2x + 3$, calculate y when $x = 0$ and then when $x = 2$.

   Ans. $y = 3$ then $y = -1$.

2. Core · Method [REG] Give two points that allow you to draw the line $2x + y - 4 = 0$.

   Ans. For $x = 0$: $y = 4$; for $x = 2$: $y = 0$. Points $(0,4)$ and $(2,0)$.

3. Core + · Quick [GEO] What is the line with equation $x = 3$? Give a direction vector. Does it have a gradient?

   Ans. The vertical line through $(3,0)$; $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ is a direction vector; it has no gradient; Cartesian equation $x - 3 = 0$.
   Cannot be written $y = mx + p$.

4. Review · Quick [CMP] Compare $\frac{3}{7}$ and $\frac{4}{9}$.

   Ans. Cross-multiplying: $3 \times 9 = 27 < 28 = 4 \times 7$, so $\frac{3}{7} < \frac{4}{9}$.

5. Thread · Quick [EQ] Solve in $\mathbb{R}$: $5x - 7 = 2x + 8$.

   Ans. $x = 5$
   Pitfall: collect the x terms on one side, changing signs correctly.

Sheet 9 Tue 10 Nov. 2026 *Finding the equation of a line: two points, a point and a direction vector, a point and a gradient*

1. Warm-up · Quick [GEO] Gradient of the line through $(1,3)$ and $(3,7)$?

   Ans. $\frac{7-3}{3-1} = 2$.

2. Core · Method [GEO] Slope-intercept equation of the line with gradient 2 through $A(1,3)$.

   Ans. $y - 3 = 2(x - 1)$, that is $y = 2x + 1$.
   Check: $2 \times 1 + 1 = 3$.

3. Core + · Method [GEO] Line through $A(1,2)$ with direction vector $\vec{u}\begin{pmatrix}2\\-1\end{pmatrix}$: Cartesian equation using the determinant of $\overrightarrow{AM}$ and $\vec{u}$.
   Ans. $M(x,y) \in d \Leftrightarrow \det\left(\overrightarrow{AM},\vec{u}\right) = 0$: $(x-1)(-1)-(y-2)\times 2 = 0$, that is $x+2y-5=0$.
   Check: $1+4-5=0$; $\begin{pmatrix}-2\\1\end{pmatrix}$ *is indeed collinear to* $\begin{pmatrix}2\\-1\end{pmatrix}$.
4. Review · Quick [PUI] Simplify $2^5 \times 2^{-3}$.
   Ans. $2^{5-3} = 2^2 = 4$.
5. Thread · Table → formula [REG] An affine function f satisfies: $f(0) = -1$, $f(1) = 1$, $f(2) = 3$. Give $f(x)$.
   Ans. Gradient 2, y-intercept -1: $f(x) = 2x - 1$.

Sheet 10 Thu 12 Nov. 2026 *Lines and affine functions: linking the representations*

1. Warm-up · Quick [IMG] $f(x) = -3x + 4$. Calculate $f(2)$ and the preimage of 10.
   Ans. $f(2) = -2$; $-3x+4 = 10 \Leftrightarrow x = -2$.
2. Core · Method [REG] Find the affine function f such that $f(1) = 1$ and $f(3) = 5$.
   Ans. Gradient $\dfrac{5-1}{3-1} = 2$; $f(x) = 2x + p$ with $2+p = 1$: $f(x) = 2x - 1$.
3. Core + · True/false [REG] True or false? Justify. Every line in the plane is the graph of an affine function.
   Ans. False. Counter-example: the vertical line $x = 3$ is not the graph of any function (the number 3 would have infinitely many images).
4. Review · Quick [PRB] A fair 6-sided die is rolled. What is the probability of getting a multiple of 3?
   Ans. Outcomes 3 and 6: $\dfrac{2}{6} = \dfrac{1}{3}$.
5. Thread · Quick [GEO] ABC is right-angled at A with $AB = 6$ cm and $AC = 8$ cm. Calculate BC.
   Ans. $BC^2 = 36 + 64 = 100$, so $BC = 10$ cm.

Sheet 11 Mon 16 Nov. 2026 *Collinearity, parallelism, intersection of lines*

1. Warm-up · Quick [SUB] Does the point $(3,5)$ lie on the line $y = 2x - 1$?
   Ans. $2 \times 3 - 1 = 5$: yes.
2. Core · Method [GEO] Relative positions of $d:\ y = 3x + 1$ with $d':\ y = 3x - 2$, then with $d'':\ y = -x + 1$.
   Ans. d and d': same gradient, different y-intercepts: strictly parallel; d and d'': different gradients: intersecting.
3. Core + · Method [GEO] Show in two ways that $A(1,4)$, $B(2,7)$ and $C(4,13)$ are collinear.
   Ans. (i) All three satisfy $y = 3x + 1$. (ii) $\overrightarrow{AB}\begin{pmatrix}1\\3\end{pmatrix}$, $\overrightarrow{AC}\begin{pmatrix}3\\9\end{pmatrix}$: $\det\left(\overrightarrow{AB},\overrightarrow{AC}\right) = 9 - 9 = 0$.
4. Review · Quick [PRO] On a map with scale $\dfrac{1}{50{,}000}$, two towns are 4 cm apart. What is the real distance?
   Ans. $4 \times 50{,}000 = 200{,}000$ cm $= 2$ km.
5. Thread · Method [ALG] Programme: choose a number x; add 3; multiply by 2; subtract 6. Calculate the result for $x = 5$, then write an expression in x and simplify it. What do you notice?
   Ans. For 5: 10. Expression: $2(x+3) - 6 = 2x$: the result is always twice the chosen number.
   Link A09: algebraic notation proves what numerical trials suggest.

Sheet 12 Tue 17 Nov. 2026 *Intersection of two lines; Python collinearity function*

1. Warm-up · Quick [EQ] Solve $2x + 1 = -x + 7$.

   Ans. $3x = 6$, so $x = 2$.

2. Core · Method [GEO] Coordinates of the intersection of $y = 2x + 1$ and $y = -x + 7$.

   Ans. $x = 2$ then $y = 2 \times 2 + 1 = 5$: $(2, 5)$.

   Check in the second equation: $-2 + 7 = 5$.

3. Core + · Code reading [ALG] What do `alignes(0,0,1,2,2,4)` and `alignes(0,0,1,2,2,5)` return?

   ```
   def alignes(xa, ya, xb, yb, xc, yc):
       return (xb-xa)*(yc-ya) - (yb-ya)*(xc-xa) == 0
   ```

   Ans. `True` then `False`: determinants $1 \times 4 - 2 \times 2 = 0$ and $1 \times 5 - 2 \times 2 = 1 \neq 0$.

4. Review · Expand [DEV] Expand and simplify $3(x - 2)$.

   Ans. $3x - 6$.

5. Thread · Prove [LIT] Show that the sum of three consecutive integers is a multiple of 3.

   Ans. With $n - 1,\ n,\ n + 1$: sum $= 3n$, a multiple of 3.

   Choosing the middle term simplifies the calculation.